

House Price Prediction - Project Summary

The objective of this project was to predict house prices using machine learning techniques. I used the Housing.csv dataset and performed data loading, exploration, cleaning, and preprocessing before building the models.

During data exploration, I found that the dataset contained 545 records and there were no missing values. Categorical columns were converted into numerical form using one-hot encoding so that machine learning models could work with the data.

For model building, I trained two models: Linear Regression and Random Forest Regressor. After evaluating both models, Linear Regression performed better with an R^2 score of approximately 0.65, while Random Forest achieved an R^2 score of around 0.61.

I also created visualizations such as a house price distribution histogram, a correlation heatmap, and an actual vs predicted price chart. These helped in understanding the data and model performance.

From the analysis, I observed that features such as area, bathrooms, stories, and other house amenities had a noticeable impact on house prices. One interesting finding was that Linear Regression performed better than Random Forest, which was unexpected.

Overall, this project helped me understand the complete machine learning workflow, including data preprocessing, model training, evaluation, and visualization.

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